A TENSOR-SQUARE FLAT ALGEBRA OVER A FIELD IS WEAKLY ETALE AND A FILTERED COLIMIT OF ETALE ALGEBRAS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot; nothing was added, removed or redefined and no package was added.

Editorial note. The source URL is the source-level Stacks Project URL, as required when more than one card is produced from one source; the per-claim tag is carried by claim_key.

Editorial note. The author writes 'hence B is a absolutely flat ring' and 'a finitely generated K-sub algebra'; both phrasings are reproduced unchanged. The nine references inside the counted proof span are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- Absolutely flat rings: reduced, with every local ring a field. invoked at proof lines 30522-30525. Gives reducedness of B and the field property of its local rings, which the rest of the argument uses.
- Weakly etale maps and their composition, and weakly etale fields as separable algebraic extensions. used at proof lines 30528-30532. Converts the hypothesis into the separability of the residue field extensions.
- Dimension-zero rings as products of their local rings, and etale algebras over a field as reduced rings with finite separable residue fields. used at proof lines 30546-30552. Is the criterion that turns the dimension-zero conclusion into etaleness of A over K.

Source reference. Original lemma 00U3: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1097–1097. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 00GA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 600–600. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 00FK: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 574–574. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 00KJ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 753–753. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 30250: sources/stacks-more-algebra/pinned-original-source.tex, source lines 30250–30250. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 30488: sources/stacks-more-algebra/pinned-original-source.tex, source lines 30488–30488. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 30323: sources/stacks-more-algebra/pinned-original-source.tex, source lines 30323–30323. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 30347: sources/stacks-more-algebra/pinned-original-source.tex, source lines 30347–30347. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 30454: sources/stacks-more-algebra/pinned-original-source.tex, source lines 30454–30454. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let B be an algebra over a field K. The following are equivalent*

1. *$B \otimes_K B \to B$ is flat,*
2. *$K \to B$ is weakly étale, and*
3. *B is a filtered colimit of étale K-algebras.*

Moreover, every finitely generated K-subalgebra of B is étale over K.

Proof. Parts (1) and (2) are equivalent because every K-algebra is flat over K. Part (3) implies (1) and (2) by Lemma 30454

Assume (1) and (2) hold. We will prove (3) and the finite statement of the lemma. A field is absolutely flat ring, hence B is a absolutely flat ring by Lemma 30323. Hence B is reduced and every local ring is a field, see Lemma 30250.

Let $\mathfrak{q} \subset B$ be a prime. The ring map $B \to B_\mathfrak{q}$ is weakly étale, hence $B_\mathfrak{q}$ is weakly étale over K (Lemma 30347). Thus $B_\mathfrak{q}$ is a separable algebraic extension of K by Lemma 30488.

Let $K \subset A \subset B$ be a finitely generated K-sub algebra. We will show that A is étale over K which will finish the proof of the lemma. Then every minimal prime $\mathfrak{p} \subset A$ is the image of a prime $\mathfrak{q}$ of B, see Algebra, Lemma 00FK. Thus $\kappa(\mathfrak{p})$ as a subfield of $B_\mathfrak{q} = \kappa(\mathfrak{q})$ is separable algebraic over K. Hence every generic point of $\mathrm{Spec}(A)$ is closed (Algebra, Lemma 00GA). Thus $\dim(A) = 0$. Then A is the product of its local rings, e.g., by Algebra, Proposition 00KJ. Moreover, since A is reduced, all local rings are equal to their residue fields which are finite separable over K. This means that A is étale over K by Algebra, Lemma 00U3 and finishes the proof. □